



Study: **DMR-QA 45**
NPDES Permit #: **MI0020257**
Laboratory Name: **Cadillac WWTP**

Inorganic Results





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DMR-QA 45 Final Report

NPDES Permit #: MI0020257
Permit Holder: Cindy A Tomaszewski
 Lab Supervisor
 Cadillac WWTP
 1121 Plett Rd.
 Cadillac, MI 49601
 (231) 775-2368

ERA Customer Number: C454401
Report Issued: 09/26/2025
Study Dates: 06/06/2025 - 08/29/2025

TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
WP Minerals (cat# 581, lot# P364-506)											
1610	Conductivity at 25°C	µmhos/cm	Acceptable	318	313	282 - 344	SM 2510 B-2011 2011	313	6.94	MI01099	WP364
1950	Total Solids at 105°C	mg/L	Not Acceptable	749	209	164 - 254	SM 2540 G-2011 2011	215	14.0	MI01099	WP364



Study: **DMR-QA 45**
NPDES Permit #: **MI0020257**
Laboratory Name: **Cadillac WWTP**

Microbiology Results



A Waters Company

DMR-QA 45 Final Report

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 Cadillac, MI 49601
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TNI Analyte Code	Analyte	Units	Performance Evaluation	Reported Value	Assigned Value	Acceptance Limits	Method Description	Study Mean	Study Standard Deviation	USEPA Lab Code	Study
DMRQA WasteWatR™ Coliform MicrobE™ (cat# 576, lot# Q045-083)											
2530	Fecal Coliform (MPN-Multiple Well)	MPN/100mL	Acceptable	186.0	183	68.0 - 492	ColiAlert®-18 (Fecal Coliforms) 2010	183	70.6	MI00082	DMRQA45





June 30, 2025

Claire Handley
Michigan (WP)
MI EGLE - Water Resources Division
Constitution Hall
525 West Allegan Street
Lansing, MI 48933

Enclosed are the final reports for ERA's WP-364 WatR™ Pollution Proficiency Testing (PT) study. Attached for your convenience is a table of all of the laboratories that participated in this WatR™ Pollution study for your certification.

All analytes in ERA's WP-364 WatR™ Pollution Proficiency Testing study have been evaluated using the following tiered approach. If the analyte is listed in the most current TNI Fields of Proficiency Testing (FoPT) table the evaluation was completed by comparing the reported result to the acceptance limits generated using the criteria contained in the table and the evaluation criteria contained in the 2016 TNI Standard, Volume 3. If the analyte is not included in the TNI FoPT table, the reported result has been evaluated using the procedures outlined in ERA's Standard Operating Procedure for the Generation of Performance Acceptance Limits (SOP 730002268).

If you have any questions regarding ERA's WatR™ Pollution Proficiency Testing program, please contact our Proficiency Testing Department at 1-800-372-0122.

Sincerely,

Craig Huff
Senior Technical Manager

attachments



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Laboratory	Contact/Phone Number	State ID	EPA ID
Cadillac WWTP	Cindy A Tomaszewski / (231) 775-2368	EPA MI01099	MI01099
Paragon Laboratories, Inc.	Paul Kijek / (734) 469-5626	9901	MI01187



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WP-364 Definitions & Study Discussion

Study Dates: 05/12/2025 - 06/26/2025

Report Issued: 06/30/2025

WP Study Definitions

The Reported Value is the value that the laboratory reported to ERA.

The ERA Assigned Values are compliant with the most current TNI Fields of Proficiency Testing (FoPT) table. A parameter not added to the standard is given an Assigned Value of "< PTRL" per the guidelines contained in the 2016 TNI Standard. The assigned values are directly traceable to the commercially prepared starting materials used to manufacture the PT standards.

The Acceptance Limits are established per the criteria contained in the most current TNI FoPT table or ERA's SOP for the Generation of Performance Acceptance Limits™ as applicable.

The Performance Evaluation:

Acceptable	= Reported Value falls within the Acceptance Limits.
Not Acceptable	= Reported Value falls outside the Acceptance Limits.
No Evaluation	= Reported Value cannot be evaluated.
Not Reported	= No Value reported.

The Method Description is the method the laboratory reported to ERA.

WP Study Discussion

ERA's WP-364 WatR™Pollution Proficiency Testing study has been reviewed by ERA senior management and certified compliant with the requirements of the 2016 TNI Standard and the criteria contained in the most current TNI Fields of Proficiency Testing (FoPT) table.

ERA's WP-364 WatR™Pollution study standards were examined for any anomalies. A full review of all homogeneity, stability and accuracy verification data was completed. All analytical verification data for all analytes met the acceptance criteria contained in the 2016 TNI Standard and the criteria contained in the most current TNI FoPT table.

All activities associated with this proficiency testing study were performed by Waters/ERA with the exception of those noted below. The following physical samples/products were manufactured for Waters/ERA by a subcontractor:

Microbiology products with the following catalog numbers:
880, 935, 079, 077, 080, 595, 595A, 576, 576A

The data submitted by participating laboratories was also examined for study anomalies. There were no anomalies observed during the statistical review of the data.

ERA's WP-364 WatR™Pollution study reports shall not be reproduced except in their entirety and not without the permission of the participating laboratories. The report must not be used by the participating laboratories to claim product endorsement by any agency of the U. S. government.

The data contained herein are confidential and intended for your use only.

If you have any questions or concerns regarding your assessment in ERA's WatR™Pollution Proficiency Testing program, please contact our Proficiency Testing Department at 1-800-372-0122.





Final Report Results For Laboratory Cadillac WWTP

All analytes except for 1,1-Biphenyl, 1,2-Diphenylhydrazine, 1,3-Dinitrobenzene, 2,3-Dichloroaniline, Acetophenone, Atrazine, Azobenzene, Benzaldehyde, Caprolactam, n-Decane and n-Octadecane in the Base/Neutrals, catalog # 833, are included in ERA's A2LA accreditation. Lab Code 1539.01.

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Final Evaluation Report

Study: **WP-364**

ERA Customer Number: **C454401**

Laboratory Name: **Cadillac WWTP**

Inorganic Results



All analytes except for 1,1-Biphenyl, 1,2-Diphenylhydrazine, 1,3-Dinitrobenzene, 2,3-Dichloroaniline, Acetophenone, Atrazine, Azobenzene, Benzaldehyde, Caprolactam, n-Decane and n-Octadecane in the Base/Neutrals, catalog # 833, are included in ERA's A2LA accreditation. Lab Code 1539.01.

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A Waters Company

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Lab Supervisor
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EPA ID:
ERA Customer Number:
Report Issued:
Study Dates:
Agency ID:

MI01099
C454401
06/30/2025
05/12/2025 - 06/26/2025
EPA MI01099

TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
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WP Minerals (cat# 581, lot# P364-506)

1505	Alkalinity as CaCO ₃	mg/L		40.6	34.5 - 46.7	Not Reported				40.3	2.61	
1575	Chloride	mg/L		51.0	44.2 - 57.9	Not Reported				52.3	2.56	
1610	Conductivity at 25°C	µmhos/cm	318	313	282 - 344	Acceptable	SM 2510 B-2011 2011	6/24/2025	0.711	313	6.94	Taysha Laczko
1730	Fluoride	mg/L		2.68	2.15 - 3.10	Not Reported				2.68	0.165	
1125	Potassium	mg/L		19.4	15.5 - 23.3	Not Reported				19.4	1.07	
1155	Sodium	mg/L		50.8	40.6 - 61.0	Not Reported				50.5	2.82	
2000	Sulfate	mg/L		17.1	13.3 - 20.1	Not Reported				17.1	1.01	
1955	Total Dissolved Solids at 180°C	mg/L		207	162 - 252	Not Reported				201	17.8	
1950	Total Solids at 105°C	mg/L	749	209	164 - 254	Not Acceptable	SM 2540 G-2011 2011	6/24/2025	38.2	215	14.0	Taysha Laczko

All analytes except for 1,1-Biphenyl, 1,2-Diphenylhydrazine, 1,3-Dinitrobenzene, 2,3-Dichloroaniline, Acetophenone, Atrazine, Azobenzene, Benzaldehyde, Caprolactam, n-Decane and n-Octadecane in the Base/Neutrals, catalog # 833, are included in ERA's A2LA accreditation. Lab Code 1539.01.

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WP-364 Laboratory Exception Report

Cindy A Tomaszewski
Lab Supervisor
Cadillac WWTP
200 N Lake St
Cadillac, MI 49601
(231) 775-2368

EPA ID: MI01099
ERA Customer Number: C454401
Report Issued: 06/30/2025
Study Dates: 05/12/2025 - 06/26/2025
Agency ID:

Not Acceptable Evaluations

TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description
WP Minerals (cat# 581, lot# P364-506)							
1950	Total Solids at 105°C	mg/L	749	209	164 - 254	Not Acceptable	SM 2540 G-2011 2011



All analytes except for 1,1-Biphenyl, 1,2-Diphenylhydrazine, 1,3-Dinitrobenzene, 2,3-Dichloroaniline, Acetophenone, Atrazine, Azobenzene, Benzaldehyde, Caprolactam, n-Decane and n-Octadecane in the Base/Neutrals, catalog # 833, are included in ERA's A2LA accreditation. Lab Code 1539.01.

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Study # : WP-364

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NPDES Number: MI0020257

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DMRQA45 Graded Results Report

Study: WP0425 for DMRQA45

Opening Date: April 1, 2025 - Closing Date: May 15, 2025

EPA Lab ID: MI00106

NPDES Permit ID: MI0020257

Laboratory: Trace Analytical Labs
2241 Black Creek Rd.
Muskegon, MI 49444
USA
Ms. Alyson Yagiela
231-773-5998

Permittee Name: CADILLAC WWTP

Trace Metals 1 (PT-TM1-WP)

Lot #: 8265-04

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
1010	Arsenic	10014605	EPA 200.8 Rev.5.4 1994	µg/L	165	170	128 - 201	Acceptable
1020	Beryllium	10014605	EPA 200.8 Rev.5.4 1994	µg/L	259	275	220 - 297	Acceptable
1030	Cadmium	10014605	EPA 200.8 Rev.5.4 1994	µg/L	136	138	115 - 156	Acceptable
1055	Copper	10014605	EPA 200.8 Rev.5.4 1994	µg/L	250	270	212 - 287	Acceptable
1150	Silver	10014605	EPA 200.8 Rev.5.4 1994	µg/L	491	508	418 - 565	Acceptable

Mercury (PT-HG-WP)

Lot #: 8265-05

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
1095	Mercury	10036609	EPA 245.1 Rev.3 1994	µg/L	21.5	16.7	15.1 - 28.0	Acceptable

Minerals 1 (PT-MIN1-WP)

Lot #: 8265-08

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
1575	Chloride	10053200	EPA 300.0 Rev.2.1 1993	mg/L	196	214	173 - 220	Acceptable
2000	Sulfate	10053200	EPA 300.0 Rev.2.1 1993	mg/L	8.26	8.77	5.78 - 10.1	Acceptable

Mercury Low Level (PT-HGLL-WP)

Lot #: 8265-47

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
1095	Mercury	10237204	EPA 1631E 2004	ng/L	59.5	56.4	43.7 - 74.6	Acceptable

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DMRQA45 Graded Results Report

Study: DMRQA45-WET

Opening Date: June 6, 2025 - Closing Date: August 29, 2025

EPA Lab ID: MI01182

NPDES Permit ID: MI0020257

Laboratory: Global Environmental Consulting
223 W. Michigan Ave.
Clinton, MI 49236
USA
Mr. David Cook
7346492998

Permittee Name: CADILLAC WWTP

Fathead Minnow Method 13 (PT-13-WET)

Lot #: 8565-13

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
3410	Fathead Minnow Acute MHSF 25° - LC50	3410	EPA 2000.0 - Fathead minnow, 48-hr Acute, nonrenewal, MHSF 25°C 2002	S.U.	35.8	37.5	29.8 - 41.9	Acceptable

Fathead Minnow Method 15 (PT-15-WET)

Lot #: 8565-15

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
3410	Fathead Minnow Chronic MHSF - Survival NOEC	3410	EPA 1000.0 - Fathead minnow, 7-day Chronic, daily renewal, MHSF 25°C 2002	S.U.	50	25	25 - 100	Acceptable
3410	Fathead Minnow Chronic MHSF - Growth IC25 (ON)	3410	EPA 1000.0 - Fathead minnow, 7-day Chronic, daily renewal, MHSF 25°C 2002	S.U.	47.2	55.98	23.4 - 71.0	Acceptable
3410	Fathead Minnow Chronic MHSF - Growth NOEC (ON)	3410	EPA 1000.0 - Fathead minnow, 7-day Chronic, daily renewal, MHSF 25°C 2002	S.U.	25	25	12.5 - 50	Acceptable

Ceriodaphnia Method 19 (PT-19-WET)

Lot #: 8565-19

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
3315	Ceriodaphnia Acute MHSF 25° - LC50	3315	2002.0	S.U.	42.0	33.97	21.1 - 63.0	Acceptable

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DMRQA45 Graded Results Report

Study: DMRQA45-WET

Opening Date: June 6, 2025 - Closing Date: August 29, 2025

EPA Lab ID: MI01182

NPDES Permit ID: MI0020257

Laboratory: Global Environmental Consulting
223 W. Michigan Ave.
Clinton, MI 49236
USA
Mr. David Cook
7346492998

Permittee Name: CADILLAC WWTP

Ceriodaphnia Method 21 (PT-21-WET)

Lot #: 8565-21

NELAC Code	Analyte	Method Code	Method Description	Units	Assigned Value	Result	Acceptance Limits	Evaluation
3315	Ceriodaphnia Chronic MHSF - Survival NOEC	3315	1002.0	S.U.	25	50.0	12.5 - 50	Acceptable
3315	Ceriodaphnia Chronic MHSF - Reproduction IC25	3315	1002.0	S.U.	29.4	28.01	23.1 - 35.8	Acceptable
3315	Ceriodaphnia Chronic MHSF - Reproduction NOEC	3315	1002.0	S.U.	25	25	12.5 - 50	Acceptable



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Final Evaluation Report

Study: WP-363

ERA Customer Number: C454401

Laboratory Name: Cadillac WWTP

Inorganic Results



All analytes except for 1,1-Biphenyl, 1,2-Diphenylhydrazine, 1,3-Dinitrobenzene, 2,3-Dichloroaniline, Acetophenone, Atrazine, Azobenzene, Benzaldehyde, Caprolactam, n-Decane and n-Octadecane in the Base/Neutrals, catalog # 833, are included in ERA's A2LA accreditation. Lab Code 1539.01.
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Study #: WP-363





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WP-363 Final Evaluation Report

Ver. 1
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EPA ID:
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Report Issued:
Study Dates:
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MI01099
C454401
06/02/2025
04/14/2025 - 05/29/2025
EPA MI01099

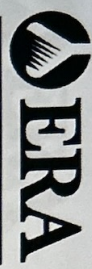
TNI Analyte Code	Analyte	Units	Reported Value	Assigned Value	Acceptance Limits	Performance Evaluation	Method Description	Analysis Date	Z Score	Study Mean	Study Standard Deviation	Analyst Name
WP Dissolved Oxygen (cat# 212, lot# P363-213)												
1880	Oxygen, Dissolved	mg/L	8.18	7.94	5.56 - 10.3	Acceptable	HACH 10360 1.2.2011	5/28/2025	0.304	8.06	0.391	Taysha Lazko



All analytes except for 1,1-Biphenyl, 1,2-Diphenylhydrazine, 1,3-Dinitrobenzene, 2,3-Dichloroaniline, Acetophenone, Atrazine, Azobenzene, Benzaldehyde, Caprolactam, n-Decane and n-Octadecane in the Base/Neutrals, catalog # 833, are included in ERA's A2LA accreditation. Lab Code 1539.01.
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WP-363 Laboratory Exception Report

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(231) 775-2368

EPA ID:
ERA Customer Number:
Report Issued:
Study Dates:
Agency ID:

MI01099
C454401
06/02/2025
04/14/2025 - 05/29/2025

Not Acceptable Evaluations

There were no Not Acceptable evaluations for this study.



All analytes except for 1,1-Biphenyl, 1,2-Diphenylhydrazine, 1,3-Dinitrobenzene, 2,3-Dichloroaniline, Acetophenone, Atrazine, Azobenzene, Benzaldehyde, Caprolactam, n-Decane and n-Octadecane in the Base/Neutrals, catalog # 833, are included in ERA's A2LA accreditation. Lab Code 1539.01.
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Study #: WP-363

